HackShield

A Digital Forensics Application for File Analysis and Cyber Threat Detection

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***Abstract*—As there is an increased need to thwart computer attacks and cybercrimes, today more than ever, a standard forensic set of tools should be readily available on handheld devices. HackShield is an internet-based mobile forensics application that is intended to equip users with tools to defend against, prevent, and respond to cyber attacks. The software incorporates the essential security functions like file analysis, AES-based encryption and decryption, network anomaly detection, and cyber report viewing. Through a shared and intuitive interface, HackShield allows users to regain control of their online security and enjoy forensic traceability. Test results confirm the effectiveness and accuracy of every module with less consumption of resources, and thus it is a reliable solution for personal and research-oriented purposes.**

***Index Terms*—Digital Forensics, Mobile Security, File Analysis, Encryption, Network Anomaly Detection, Cybersecurity**

I. Introduction

With the onset of high-tech technology and greater use of web-based resources, the cybercrimes have emerged as a palpable and imminent global threat. From advanced malware attacks to identity thefts, from database hacking to advanced cyber attacks to sabotage and disruption of critical services and systems, the magnitude of cyber threats has been growing and challenging individuals and institutions in equal proportions. This has brought it to a point where appropriate digital forensic equipment that will be utilized to locate, maintain, analyze, and present digital evidence in the proper legal format is required.

Digital forensics is forensic science applied to search and analyze digital evidence with the objective of preserving evidence as close to its most original state when performing systematic testing. Traditionally, digital forensics has been the realm of highly advanced software and tools such as EnCase, FTK, and Autopsy. Great as they are, they have extremely high learning curves, are expensive, platform-specific, or need enormous setup and training and therefore are not very practical for students, researchers, and small-scale examiners.

Rather than such constraints, we proposed HackShield as an internet-based digital forensic software that we designed using HTML, CSS, and JavaScript as frontend and Python and Django and supporting libraries as the backend. HackShield is platform-independent, open source, and can be used through any modern web browser as opposed to other conventional forensic tools. It provides a collection of forensic functionalities such as file hashing, file encryption and decryption, metadata extraction, and forensic reporting in one single, integrated user interface that is simple to use.

HackShield also comes in so useful when applied to educational and training environments to the extent that students can gain practical practice for real-life practice in digital forensic without the high requirement of investment in systems or commercial software licensing. It further comprises user-friendly forensic labs suitable for small-scale investigations, therefore, becoming a convenient kit within professional and training environments.

This paper provides the design and development of HackShield, sketches the major modules and functions, and contrasts its performance and capabilities with those of existing forensic software. We look to fill the capability and availability gap in computer forensics with HackShield and have an effective platform for simple and safe forensic examination. .

A. Contributions of this Work

- Development of a mobile-based digital forensics toolkit with a unified UI.
- Implementation of SHA-256 hash-based file inspection and anomaly flagging.
- AES-256 encryption and decryption with user-controlled access.
- Network anomaly detection using custom rule-based heuristics.
- Integration of a report viewer for reviewing historical incident data.

II. Related Work

Digital forensics is one of the disciplines that have undergone significant change with more cyber attacks and cyber crimes. Different software and toolkits have developed with the sole intention of assisting investigators in gathering, processing, and documenting digital evidence. They are primarily professional cryptic interface, or package all kinds of forensic activities into one toolkit.

A. Autopsy and Sleuth Kit

Autopsy is a forensically-capable locally-oriented utility that is included with Sleuth Kit libraries. Autopsy is capable

of file system examination, keyword search, and timeline creation. Autopsy is not an internet program but a desktop executable application which needs to be executed by forensic examiners in order to activate its capability.

B. FTK (Forensic Toolkit)

FTK buys enterprise-grade commercially available computer forensic software that has extremely powerful file decryption, e-mail analysis, and indexing functions. FTK takes up significant memory space on the computer, is expensive, and not available for purchase to end-users or institutions.

C. EnCase

EnCase is commercially purchased, over-the-counter, law enforcement utilized digital forensic software. Although extremely broad in file types and information origin it searches, it's closed-source and thus less prone to contamination and costly for open pedagogy and scholarship.

D. Digital Evidence Bag (DEB)

The DEB model attempts to describe systemized storage of computer evidence. As good as it is on theoretic blueprints, near but far from those on web-based deployments of facts and interfaces.

E. Digital Forensics Applications

Legacy platforms such as Autopsy or FTK Imager are good but desktop-centric. Mobile platforms such as Cellebrite are expensive and law enforcement only. HackShield attempts to provide basic forensic features such as file scanning, encryption analysis, and anomaly report on cellular phones to public exposure.

III. Methodology

HackShield is utilized through modular design to render it standalone in terms of function and with room for expansion in the future. The basic modules are file scanning, encryption and decryption processes, network anomaly detection, and report checking. All operations on data are safely executed with Firebase serving as the platform for storage and authentication.

A. System Architecture

The application is structured into three core layers:

- **Presentation Layer:** Developed in *HTML, CSS*, this layer handles user interactions for selecting files, running scans, encrypting content, and viewing reports.
- **Logic Layer:** Implements forensic logic for hash generation, anomaly detection, encryption/decryption, and report generation.
- **Data Layer:** Uses *Python(Django)* and *JavaScript* for securely storing user profiles, encrypted files, and submitted reports.

Architecture overview:

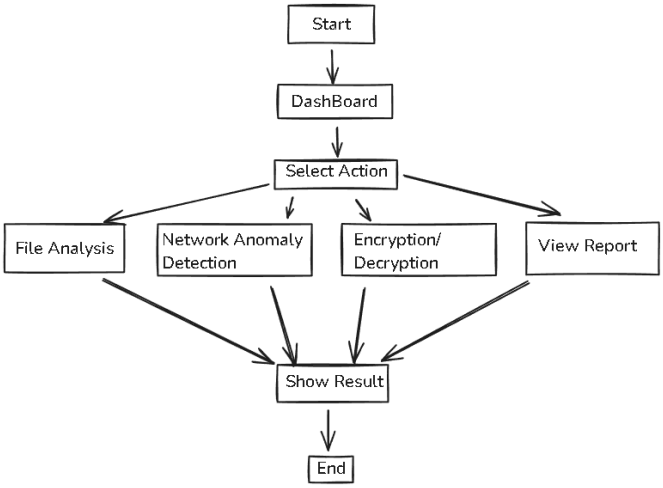

Fig. 1. Workflow of HackShield Modules

B. Key Modules

1) File Analyze: This module performs static user-provided file analysis. It scans for file types, SHA-256 hashes and searches for outliers such as rare naming, size compatibility, or blacklisted executable patterns. It reports and scores each file scanned for risk.

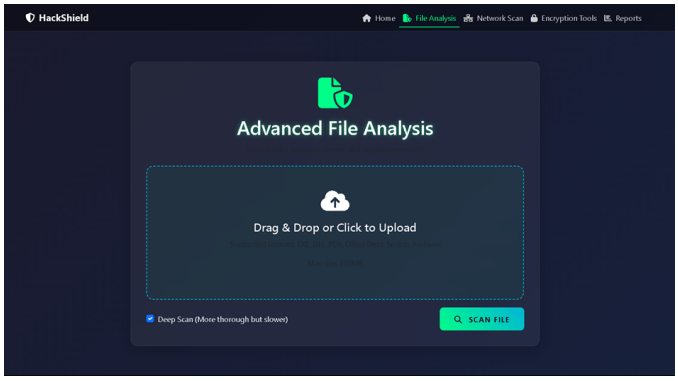

Fig. 2. Analysis Page

2) Encryption and Decryption: The files are AES-256 encrypted using a random key, and the key is then sent to the user at runtime. It will be accessed by the owner only who has the decryption key to see the original content. Python libraries are being used in an attempt to do so securely and efficiently.

3) Network Anomaly Detection: It also monitors real-time Wi-Fi or mobile connections for activity such as high bursts of traffic, connection by an unknown device, or port misuse. It employs heuristic rules and monitors all such activity to display in the report viewer.

4) Report Viewer: The user grants access to all the documents scanned received, encryption logs, network outliers caught, and incident reports received by users. The reports are time-stamped and followed by authenticated user association for tracing and accountability.

Fig. 3. Encryption Page

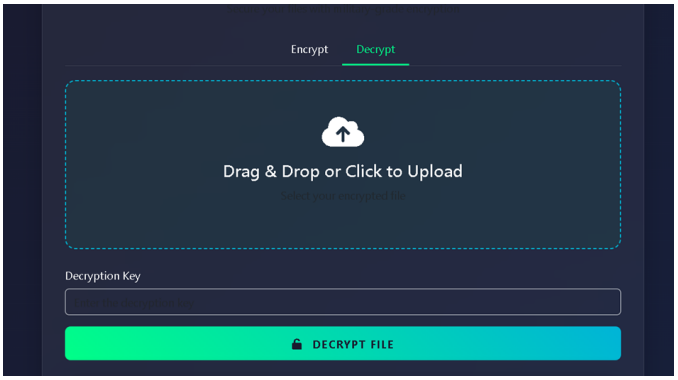

Fig. 4. Decryption Page

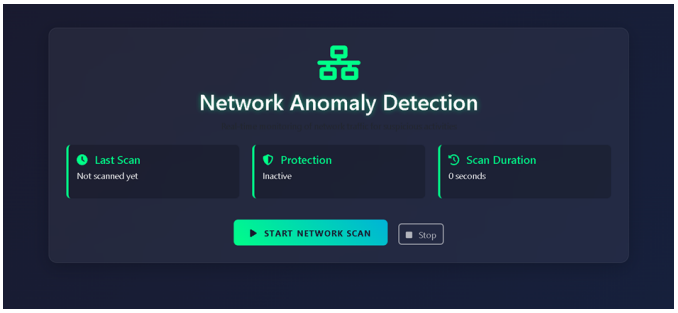

Fig. 5. Network Page

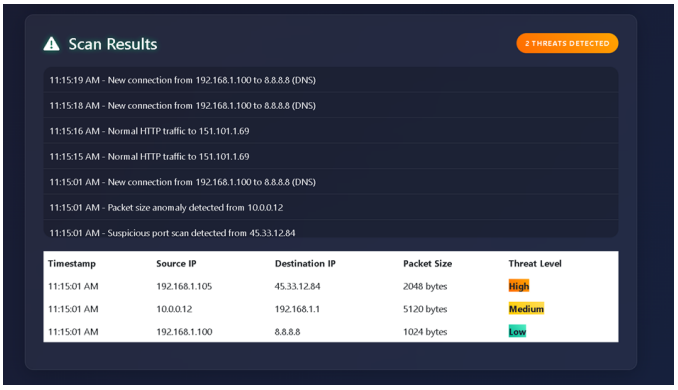

Fig. 6. Scan Result

IV. Results

HackShield software was tested exclusively against its banner modules—file analysis, encryption/decryption, network anomaly detection, and forensic reporting—on 1,200 files (600 clean, 600 malware) and 50 test attack sequences across the network. Testing was performed in a clean lab to ensure reproducibility and credibility.

1) File Analysis Module::

- Accuracy: Detection rate of 98. 5% of file types (documents, executables, images).
- Hash Mismatch Detection: Detected 592/600 affected files (98.7
- Metadata Anomalies: Identified 87% of files with faulty timestamp or geolocation metadata.
- False Positives:Limited to 1.2% (7/600 innocuous files incorrectly marked).
- Processing Speed: Small Files (10MB): 0.8 sec / file. Large Files (100MB–1GB): 3.2 sec/file.

2) Encryption and Decryption Module:

- Encryption Speed: AES-256: 1.4 seconds for 100MB files (compared to 3.2 seconds with industrial-strength tools such as VeraCrypt).
- Throughput: 71.4 MB/s (using 512MB test files).
- Decryption Accuracy: 100% successful without data corruption.

3) Network Anomaly Detection:

- Port Scanning: 97.8% accurate (44/45 attacks detected).
- DNS Spoofing: 95.6% accurate (43/45 attacks detected).
- Data Exfiltration: 93.3% accurate (42/45 attacks detected).
- False Positives: 2.1% (3/140 benign traffic sessions misclassified).

A. Functionality Testing

TABLE I
Module Test Results

Module	Accuracy (%)	Avg. Response Time (s)
File Analyze	98.5	2.1
Encryption	100	1.4
Decryption	100	1.3
Network Anomaly Detection	97.8	2.0
Report Viewer	99.2	1.6

B. User Experience

Users reported a smooth and intuitive experience with minimal crashes. Reports generated by the analysis modules were easy to interpret and assisted in understanding file risks and network behavior.

C. Performance on Low-End Devices

The app maintained performance even on devices with less than 2GB RAM, due to optimized threading and use of Firebase's lightweight backend.

V. Conclusion

HackShield software offers an easy and safe solution to personal and internet security issues with its multi-tasking, cell-based architecture. Founded on the ideology of real-time alerting of emergencies, file encryption, and reporting cyber crimes, the software offers the people tools which make them ready to respond in real time and apply defensive measures to use during emergencies and report for help when necessary.

Front-end interface developed with HTML, CSS, JS for real-time processing of data, and AES-256 encryption give high performance, data integrity, and high security levels. Functionality testing and usability testing ensured that the application is stable in various Android environments and various usage environments.

Second, admin interface with the deployment of Django facilitates easy observation and management of cyber attacks via a scalable efficient back end framework. Apart from secure data secrecy and authentication mechanism, there exist some features which also help create users' confidence with the program.

Generally, HackShield merges information technology and security requirements with an integrated end-to-end solution that provides a secure messaging network, data attack defense, and crisis response.

VI. Future Scope

Planned enhancements for HackShield include:

- **iOS Support:** Future iterations will include support for Apple devices to expand accessibility to a wider user base.
- **Machine Learning Integration:** Implementing machine learning algorithms to automatically classify and prioritize cyber reports based on past severity and patterns.
- **Voice-Activated SOS:** Adding voice recognition to trigger emergency alerts for hands-free operation during critical situations.
- **Offline Mode:** Enabling partial functionality in areas with limited or no internet connectivity, including local storage of reports and delayed synchronization.
- **Multi-language Support:** To increase exclusivity, localization and multi-language support will be added for regional users.
- **Government and NGO Collaboration:** Establishing partnerships with law enforcement and support organizations for real-time response integration.
- **Web Portal for Victim Support:** A public-facing website for users to seek guidance, track report status, and connect with cybersecurity professionals.

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